Needs analysis and a 12-week training programme for performance enhancement in a 20-year-old female wing attack netball player

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Netball is a high-intensity intermittent court sport characterised by frequent accelerations, decelerations, rapid changes of direction, jumping and landing, and sustained technical decision-making under pressure (Mackay *et al*., 2025). Match play consists of four fifteen-minute quarters, during which players perform hundreds of movement actions, including shuffling, sprinting, pivoting, and aerial contesting (World Netball, 2024).

These repeated high-impact and multidirectional demands exert significant stress on the neuromuscular, cardiovascular, and musculoskeletal systems, making netball a sport that requires a well-rounded combination of strength, power, agility, aerobic fitness, and technical skill (Young *et al*., 2016) (Downs *et al*., 2021) (Thomas *et al*., 2017) (Brooks *et al*., 2020). Simultaneously, the unpredictable nature of landings, body contact, and rapid directional changes heightens the risk of lower-limb injuries, particularly ankle sprains and anterior cruciate ligament (ACL) injuries (Mullally *et al*., 2023) (Downs *et al*., 2021) (Horne *et al*., 2024). Therefore, effective netball performance preparation must combine both performance enhancement and injury risk reduction (Lauersen *et al*., 2014).

For this assignment, the athlete is a 20-year-old female Wing Attack (WA) competing at the university level. She measures 175 cm in height and 68 kg in body mass, with a training age of approximately 8 years (having played competitively since age 12). These anthropometric characteristics align closely with normative data reported for female academy and university-level netball players, where mid-court players (including centres and wing positions) typically present with heights ranging from 170–178 cm and body masses of 65–72 kg, facilitating the agility and endurance demands of the position (Thomas *et al*., 2019) (Sinclair *et al*., 2020). Her body composition, estimated at approximately 20–22% body fat, reflects a lean yet robust build suitable for repeated high-intensity efforts, consistent with profiles of state-level and academy mid-court players (Graham *et al*., 2020). Nutritional analysis indicates a balanced diet meeting her estimated daily energy requirements (approximately 2800–3200 kcal, adjusted for training load), with adequate carbohydrate intake to support glycogen replenishment during intermittent match play and protein for recovery (Costello *et al*., 2024).

Within this context, the role of the rehabilitator is central to promoting athlete health, optimising physical performance, and reducing injury risk through structured exercise planning. A rehabilitator works across health monitoring, corrective exercise, and sports conditioning, ensuring that training loads are appropriate, movement patterns are efficient, and performance progressions are evidence-based (British Association of Sport Rehabilitators and Athletic Therapy Ltd) (Bourdon *et al*., 2017). This dual focus on performance and protection is essential in netball, where the physical demands of the sport require athletes to be highly conditioned and resilient to the stresses of high-impact and multidirectional play (Steele, 1990). The following sections will outline a needs analysis of the sport and present a 6-week performance enhancement programme tailored to the needs of the assumed athlete.

The Wing Attack (WA) is a pivotal mid-court position primarily responsible for transitioning play from defence to attack and feeding the shooting circle (World Netball, 2024). Consequently, the movement profile of a WA is characterised by high-frequency, high-intensity intermittent activity (Thomas *et al*., 2017). Unlike linear sprinting sports, a WA performs frequent multi-directional movements, including lateral shuffling, backwards running, and sharp 180-degree turns to evade tight marking (Poole *et al*., 2017). Research indicates that mid-court players cover significant distances, but more importantly, engage in a higher density of accelerations and decelerations compared to holding positions, such as the Goal Shooter (van *et al*., 2020). This places a substantial demand on the ATP-PC energy system for explosive bursts, supported by aerobic capacity for recovery between efforts (Glaister, 2005).

A critical biomechanical constraint in netball is the "footwork rule," which requires rapid deceleration to a complete stop within 1.5 steps after receiving the ball (Rules of Netball., 2020) (Clark., 2021). For a WA, who often gets the ball while sprinting towards the circle edge, this requires generating substantial eccentric braking forces which mainly from the quadriceps and gluteal musculature to dissipate forward momentum (Harper *et al*., 2022) (Zhang *et al*., 2021). These manoeuvres frequently involve unilateral landings, which generate high Vertical Ground Reaction Forces (VGRF)—often exceeding 3-4 times body weight (Ishida *et al*., 2024).

If an athlete lacks sufficient eccentric strength or neuromuscular control, these high forces cannot be absorbed effectively by the muscles (Lepley *et al*., 2017) (Harper *et al*., 2020). Instead, the load is transferred to passive structures, such as the ligaments. This mechanism, particularly when combined with dynamic valgus during pivoting or landing, is the primary factor of non-contact ACL injuries in female netballers (Renstrom *et al*., 2008).

The WA must constantly create space through agility manoeuvres. Biomechanically, effective change-of-direction performance depends on the ability to lower the Centre of Mass and apply lateral force into the ground to switch vectors rapidly (Singh *et al*., 2025). Furthermore, the unique requirement of pivoting in netball introduces significant rotational torque at the knee and ankle (Greene *et al*., 2014). The plantar foot remains fixed while the body rotates, requiring high ankle stability and hip external rotator strength to maintain alignment (Malloy *et al*., 2016). For a WA, the ability to pivot under pressure while maintaining trunk stability is essential not just for injury prevention, but for the technical execution of accurate passing into the circle (Sasaki *et al*., 2011) (Zemková and Zapletalová, 2022).

Following the needs analysis, which identified high demands on change-of-direction speed, eccentric braking during deceleration and landing, explosive lower-limb power for re-acceleration and aerial contests, repeated high-intensity efforts, and effective neuromuscular control to mitigate injury risk, an appropriate battery of field-based performance tests was selected (Zemková *et al*., 2018) (Sonesson *et al*., 2021). The chosen tests were required to be practical, time-efficient, and feasible within a university training environment, while also demonstrating acceptable reliability and validity in female team-sport athletes, including netball players (Mungovan *et al*., 2018) (Osborne *et al*., 2025). Collectively, these cluster tests were designed to directly assess the key physiological and biomechanical characteristics underpinning performance in the Wing Attack position (Dobbin *et al*., 2018) (McKenzie *et al*., 2020).

The 505 Change of Direction Test was selected to evaluate the athlete's ability to decelerate rapidly and re-accelerate following a 180° change of direction (Topend Sports, n.d.). This movement pattern closely reflects the demands imposed by netball's footwork rule, where players must effectively brake, pivot, and accelerate to create space (Barber *et al*., 2016). The test involves a 10 m sprint, a plant and a 180° turn on the dominant limb, followed by a 5 m sprint back through timing gates (Jones, 2023). Previous research has demonstrated high test–retest reliability of the 505 in female netball players, supporting its sensitivity to detecting differences in COD performance (Barber *et al*., 2016).

Lower-limb explosive power was assessed using the Countermovement Jump. Explosive power and efficient utilisation of the stretch–shortening cycle are critical for repeated accelerations, rapid changes of direction, and aerial contests common to the Wing Attack role (Oleksy *et al*., 2024) (Suarez-Arrones *et al*., 2020). The CMJ was performed with hands on hips to minimise upper-body contribution and isolate lower-limb power output. Jump height was recorded using a contact mat or jump-and-reach method (Heishman *et al*., 2020) (Science for Sport, n.d.). The CMJ is widely recognised as a reliable and valid indicator of lower-body power in female athletes and has demonstrated strong associations with netball-specific performance measures (Fristrup *et al*., 2024).

The athlete’s ability to sustain repeated high-intensity efforts was evaluated using a Repeated Sprint Ability test, which consisted of six 30-m maximal sprints separated by 20 s of passive recovery (Dawson., 2012). This protocol reflects the intermittent nature of mid-court netball play, in which Wing Attack players must perform frequent accelerations and decelerations with limited recovery time (Wood *et al*., 2022) (King *et al*., 2020). Performance outcomes included total sprint time, mean sprint time, and fatigue index. RSA testing has demonstrated acceptable reliability in team-sport populations and has been shown to relate closely to the match demands observed in competitive netball (Spencer *et al*., 2006).

High-intensity intermittent aerobic capacity was assessed using the Yo-Yo Intermittent Recovery Test Level 1 (Bangsbo *et al*., 2008). This test evaluates the athlete’s ability to repeatedly perform intense shuttle running with brief recovery periods, which is essential for maintaining performance across four quarters of match play (Bangsbo *et al*., 2008). The YYIR1 consists of progressive 20 m shuttle runs interspersed with 10 s of active recovery and continues until volitional exhaustion (Schmitz *et al*., 2018). The test is commonly employed in netball research, demonstrates strong reliability, and has been shown to predict match-running performance in female players (Ben *et al*., 2020) (Graham, Scott Phillip, 2019).

Finally, neuromuscular control and landing mechanics were assessed using the Tuck Jump Assessment. Given the high incidence of non-contact anterior cruciate ligament injuries in female netball players, screening for movement deficits such as dynamic knee valgus, asymmetrical limb loading, and poor trunk control is clinically relevant (Myer *et al*., 2008). The TJA involves repeated tuck jumps over a 10 s period and is scored across 10 criteria related to lower-limb and trunk mechanics (Chimera *et al*., 2016) (Racine *et* *al*., 2021). When standardised, the TJA has demonstrated good intra- and inter-rater reliability and is recommended for identifying modifiable injury risk factors in female athletes (Myer *et al*., 2008).

Taken together, the selected testing battery enabled a comprehensive evaluation of the athlete's performance characteristics relative to the positional demands of the Wing Attack role.

The 20-year-old female Wing Attack (WA) player completed all tests after a thorough warm-up and familiarisation trials. Her results are presented in Table 1, alongside normative data derived from female netball players at academy, state, or university levels. Normative values are provided for contextual comparison and may vary according to competitive level, playing position, and specific testing protocols.

Test	Result	Normative Data (Mean ± SD or Range)	Interpretation
505 Change of Direction Test (dominant leg, seconds; lower = better)	2.68 s	2.45–2.55 s (academy/university female netballers; mid-court players often faster)	Below normative values, indicating reduced braking efficiency and re-acceleration capacity during 180° turns
Countermovement Jump (CMJ) (height, cm; higher = better)	34 cm	35–46 cm (elite/academy female netball players)	At the lower end of normative values, suggesting suboptimal lower-limb power

Test	Result	Normative Data (Mean ± SD or Range)	Interpretation
Repeated Sprint Ability (RSA) (6 × 30 m with 20 s recovery; fatigue index %, lower = better)	Fatigue index: 8.2%	4–6% (team-sport females; netball-specific protocols)	Elevated fatigue index reflects reduced ability to maintain high-intensity efforts with incomplete recovery, likely limiting sustained performance across all four quarters.
Yo-Yo Intermittent Recovery Test Level 1 (distance, m; higher = better)	1280 m (Level 17.2)	1200–1600 m (university/state-level female netball; mid-court often higher)	Within normative range, indicating adequate intermittent aerobic capacity and representing a relative strength supporting recovery between repeated efforts.
Tuck Jump Assessment (TJA) (total flaws out of 10; lower flaws= better condition)	7 flaws	<6 flaws recommended (scores ≥6 associated with elevated non-contact ACL injury risk markers)	Above the recommended threshold, demonstrating deficits in landing mechanics and neuromuscular control.

Table 1: Athlete's Test Results Compared to Normative Data for Female Netball Players

505:(Barber *et al*., 2016) (Thomas *et al*., 2019)

CMJ:(Graham *et al*., 2020) (Thomas *et al*., 2019)

RSA:(Spencer *et al*., 2005) (Bishop *et al*., 2001)

YYIRT1: (Graham *et al*., 2020) (Thomas *et al*., 2019)

TJA:(Myer *et al*., 2008)

When compared with available positional normative data, the athlete demonstrated limited lower-limb strength, suboptimal landing control, and reduced repeated-sprint capacity. These deficiencies are likely to compromise her ability to repeatedly decelerate and re-accelerate, to contest aerial balls effectively, and to maintain performance intensity over prolonged periods of match play.

The athlete demonstrates adequate intermittent aerobic recovery (Yo-Yo IR1), aligning with her ability to remain active off the ball (Barber *et al*., 2016) (Thomas *et al*., 2019). However, consistent with positional demands and performance goals for the next competitive cycle, limitations are evident in lower-limb strength and explosive power (CMJ), change-of-direction efficiency (505), repeated-sprint maintenance (RSA), and particularly landing control (TJA) (Graham *et al*., 2020) (Thomas *et al*., 2019) (Spencer *et al*., 2005) (Bishop *et al*., 2001) (Myer *et al*., 2008). These findings indicate the need to prioritise eccentric strength development, neuromuscular control, and explosive power transfer within a targeted 6-week mesocycle training programme to enhance change-of-direction speed, aerial contest performance, and high-intensity conditioning across all four quarters of match play.

To address the identified physical deficiencies among athletes, their training enhancement plan can be divided into three levels. The highest level is the macrocycle, which defines the goal for the entire training period. The next level is the mesocycle, typically spanning approximately six weeks. The smallest level is the microcycle. The smallest unit is the workout, which specifies the dosage and timing for each exercise (Lorenz *et al*., 2010).

The specific plan will be designed starting from this 6-week mesocycle (Galán-Rioja *et al*., 2023). The six-week mesocycle divides the entire training process into three consecutive 2-week blocks, each employing a distinct specialised training method: Triphasic Training, the French Contrast Method, and the Westside Conjugate Method (Sylta et al., 2016) (Elbadry *et al*.,2019) (Lorenz *et al*., 2010).

The primary objectives of this phased design are: First, through structured block periodisation, progressively stimulate and enhance all aspects of the target athlete's wing attack position—initially emphasising eccentric control and deceleration capabilities, mid phases optimising strength-to-speed conversion and power output, and finally consolidating maximum strength, dynamic explosiveness, and weakness correction (Hartmann *et al*., 2015). This ensures systematic and specific physiological adaptation.

Second, it introduces diverse training methods and exercise variations to maintain sufficient variety throughout the program. This prevents mental fatigue and diminished motivation from repetitive content, sustaining higher athlete engagement and enjoyment (Eather *et al*., 2023).

Third, this method switching mechanism incorporates a phased feedback loop. After completing each 2-week block, objective tests (e.g., CMJ, 505 Shuttle Run, Tuck Jump Assessment) and subjective feedback (RPE, recovery status) are used to evaluate the system's effectiveness (Saw *et al*., 2016). This not only provides empirical data for skill and conditioning training in ball sports like netball—supporting academic method comparisons and advancements—but also allows coaches to dynamically adjust subsequent plans based on phase performance, enabling personalised training protocols and iterative optimisation (The specific exercise schedule and content are detailed in the appendix).

In conclusion, this assignment presented a comprehensive needs analysis and performance enhancement programme for a 20-year-old female Wing Attack netball player competing at university level. By integrating sport-specific demands, positional requirements, and individual testing outcomes, the analysis highlighted the multifactorial nature of performance in the Wing Attack role, which requires high change-of-direction ability, eccentric braking capacity, explosive lower-limb power, repeated-sprint endurance, and robust neuromuscular control.

The needs analysis demonstrated that the Wing Attack position places substantial biomechanical and physiological demands on the athlete, particularly during rapid deceleration, pivoting, and unilateral landing tasks governed by the footwork rule. These demands are closely associated with both performance outcomes and injury risk, especially non-contact ACL injuries prevalent in female netball players. The selected testing battery effectively captured these demands and revealed that, while the athlete possessed adequate intermittent aerobic capacity, clear limitations were evident in lower-limb power, braking efficiency, repeated-sprint ability, and landing mechanics.

In response to these findings, a structured 6-week mesocycle training programme was developed using a block-periodised approach. The programme systematically progressed from eccentric strength and maximal force development to strength–speed conversion, plyometric efficiency, and high-velocity movement, while concurrently addressing neuromuscular control and reducing injury risk. The inclusion of evidence-based training methods and regular re-testing ensures that adaptations are monitored objectively and that training remains responsive to the athlete's development.

Overall, this programme provides a targeted, position-specific framework that aligns physical preparation with the unique demands of the Wing Attack role. By addressing identified weaknesses while maintaining existing strengths, the proposed intervention aims to enhance on-court performance, reduce injury risk, and support sustainable athletic development. This approach highlights the critical role of the rehabilitator in bridging the gap between performance enhancement and athlete health in high-demand multidirectional sports such as netball.

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Appendix

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Appendix 1: Macrocycle（Annual Training Framework）

Periodisation Chart for Netball

	January				February				March				April					May				June				July					August				September				October					November				December				
Weeks	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52
Weeks Remain	11	10	9	8	7	6	5	4	3	2	1	0	13	12	11	10	9	8	7	6	5	4	3	2	1	0																										
Taper priority																																																				
competition												X	X	X	X	X		X	X	X	X	X	X	X	X	X																										
Event Round												1	2	3	4	5	6	7	8	9	10	11	12	13	14	Final																										
Training Phase	Preparation Phase (high volume moderate intensity											Competition Phase (reduce volume increase intensity in all areas)																			Transition								Preparation													
Strength	Muscular endurance				Muscular power							Maintain						Max				Maintain				Max													Adaptation in all muscle groups for general strength													
Aerobic	Interval Aerobic training mimics game more				Circuit Aerobic Training, mimics game with little rest- gain endurance							Maintain						Maintain				Maintain				Maintain													Continuous Aerobic training to increase general aerobic fitness													
Anaerobic (includes Speed)	Anaerobic- Speed training				Anaerobic- Speed endurance training							Speed specific to position						Maintain				Maintain				Maintain													Anaerobic threshold training													
Flexibility	Static and dynamic				Static and dynamic							Maintain to prevent injury																											Dynamic stretching to increase range of motion in all muscle groups													
Skills	Skills specific to position (GA shooting, C passing)				Skills under fatigue, endurance							Developing skills under pressure/fatigue						Maintain																					General development – consistency and correct technique of ball skills, defender/attacker skills, agility skills													
Psychology	Strategies tactics				Relaxation/imagery							Focus/optimal arousal						Anxiety and Stress management								Mental rehearsal													Goal Setting									Strategies/ tactics				

Netball's macrocycle training cycle is divided into recovery, preparation, competition and transition phases, with a monthly focus on gradually building a foundation to peak and then maintaining.

January-February (Off-season recovery): Lightweight basic to rebuild strength and aerobic endurance, restore body, review technique.

March-May (early preparation period): Gradually improve strength and jumping, introduce agility and speed foundations, and build compound endurance.

June (Preseason Peak): Peak Strength, Jumping/Agility High Volume, Competition Simulated Endurance, Preparing for Preseason.

July-September (peak of the competition period): Maintain speed and strength, agility/jumping special competition transfer, low volume of high-intensity endurance, weekly competition dominance.

October-November (late game): Injury prevention maintains explosiveness and agility, speed and endurance focus, playoff peak.

December (Transition): Fully unloaded, lightly restored all qualities, and evaluated the next cycle.

Adapted from: Gleason (n.d.)； Periodisation Chart Netball (n.d.)； Buchheit et al. (n.d.)； Turner (n.d.)

Appendix 2. Mesocycle for week 1 - week 2 & exercise dosage explanation

Week1

time/day	Mon (low limb)	Tue (upper limb)	wed	Thu (low limb)	Fri (low limb)	Sat (whole bady)	Sun
Morning	Belt Squat Eccentric	Bench Press Eccentric	Rest	Reverse Hyper Eccentric	Split Squat Eccentric	Trap Bar Deadlift Eccentric	Rest
Lunch	Agility ladder drill + passing practice	Shooting accuracy	Rest	Jump landing control	Directional change sprint	Race simulation	Rest
Night	Foam roller	Shoulder Stabilization Pull + Grip	Rest	Psoas Band Kicks	Med Ball Slams	Walk Recovery	Rest

Week2

time/day	Mon (low limb)	Tue (upper limb)	wed	Thu (low limb)	Fri (low limb)	Sat (whole bady)	Sun
Morning	Belt Squat Isometric	Bench Press Isometric	Rest	Reverse Hyper Isometric	Split Squat Isometric	Trap Bar Pause Deadlift	Rest
Lunch	Agility cone barrel	Feint shot	Rest	Vertical jump hold + landing	Sideways sprint	Mini competition	Rest

time/day	Mon (low limb)	Tue (upper limb)	wed	Thu (low limb)	Fri (low limb)	Sat (whole bady)	Sun
Night	Foam roller	Shoulder strap pull + grip of equal length	Rest	Razor Curl Isometric	Med Ball Chest Pass	Yoga stretching	Rest

Key Exercise Dosage Instructions：

Belt Squat (Eccentric/Isometric): Eccentric lowering for 5-7 seconds or bottom hold for 3 seconds, 3-4 sets x 3-4 reps, 70-85% of 1RM, rest 2-3 minutes.

Reverse Hyper: Slowly controlled eccentric or 2-second hold, 3-4 sets x 4-6 reps, moderate weight, focus on posterior chain.

Split Squat: Single-leg, 3 sets x 6 reps per leg, 6-second eccentric or hold, bodyweight or light dumbbells.

Bench Press: 5-second eccentric or 2-second bottom hold, 3 sets x 3 reps, light spotter assistance, 60-80% 1RM.

Psoas Band Kicks/Razor Curl: 3 sets x 10-12 reps per leg, hold for 2 seconds with resistance, focus on hip flexors.

Med Ball Slams/Chest Pass: 3 sets x 8-10 reps, explosive throws, 4-6kg ball.

Trap Bar Deadlift: 3 sets x 3 reps, eccentric or paused, 80% 1RM.

Adapted from: de Villarreal et al. (2009); Roig et al. (2009); Schoenfeld et al. (2015); Yoshida et al. (2024)

Appendix 3. Mesocycle for week 3 - week 4 & exercise dosage explanation

Week3

time/day	**Mon (low limb Contrast)**	**Tue (upper limb Contrast)**	**wed**	**Thu (low limb Contrast)**	**Fri (Contrast)**	**Sat (whole bady)**	**Sun**
Morning	Back Squat→ Jump Squat → Bounding → Sprint	Bench Press→ Plyo Push-up → Med Ball Overhead Throw → Overhead Slam	Rest	Trap Bar Deadlift → Broad Jump →single leg Hop → Agile cone sprint	Split Squat→ Lateral Jump → Box Jump → Direction changes sprint	Contrast	Rest
Lunch	Jump shot drill + interception	Chest pass + feint	Rest	High jump landing + pass chain	Agility Ladder + Defensive Slide	Mini race simulation	Rest
Night	Plank Jump	Shoulder strap pull + grip throw	Rest	Hip extension + calf jump rope	Med Ball Twist Throw	Light running Stretch	Rest

Week4

time/day	Mon (low limb Contrast)	Tue (upper limb Contrast)	wed	Thu (low limb Contrast)	Fri (Contrast)	Sat (whole bady)	Sun
Morning	Front Squat → Depth Jump → single leg Bound → 10m Sprint	Decline Bench → Clap Push-up → Med Ball Chest Pass → Speed Pull-up	Rest	Romanian Deadlift → Vertical Jump → Lateral Bound → Zigzag Sprint	Bulgarian Split → Hurdle Hop → Reactive Jump → multi-directional sprint	Mix Contrast (Squat + push + jump)	Rest
Lunch	Aerial catch + quick pass	Spin the shot	Rest	Defensive jump	Agility	Complete confrontation	Rest
Night	Dynamic stretching	Neck stabilization	Rest	Posterior chain foam + ankle stabilization	Core Russian rotation	Yoga flow + breathing	Rest

Key Exercise Dosage Instructions：

Back Squat/Front Squat: 85-90% 1RM, 3-4 reps, 1-2 sets; rest 20 seconds before performing Jump Squats (30% 1RM, 6 reps).

Barbell Deadlift/Romanian Deadlift: 80-85% 1RM, 3 reps; followed by Standing Long Jumps (8-10 reps). Rest 10-20 seconds between sets, 3 minutes between exercise sets.

Bench Press/Incline Press: 80% 1RM, 3 reps; followed by Explosive Push-ups/Clap Push-ups (5-8 reps).

Jump Squats/Deep Jumps/Box Jumps: 6-10 reps, bodyweight or 20-30% 1RM, emphasising explosiveness.

Lateral Jumps/Hurdle Jumps/Side Jumps: 6-8 reps per leg, performed as a continuous chain movement.

Medicine Ball Throws/Passes/Slams: 4-6kg ball, 6-10 reps per side or both sides, maximum force.

Sprinting: 10-20 meters, 4-6 sets. Full effort sprints followed by 90-second recovery.

Each contrast training session includes 1-2 rounds, lasting 30-45 minutes. Netball adaptation training emphasises jump/throw transitions. Increase intensity by 5% on Thursdays.

Adapted from: Zhu et al. (2024); Pagaduan et al. (2020); Sun et al. (2025)

Appendix 4. Mesocycle for week 5 - week 6 & exercise dosage explanation

Week5

time/day	Mon (ME)	Tue (DE)	Wed	Thu (ME)	Fri (DE)	Sat (RE/GPP)	Sun
Morning	Box Squat ME	Bench DE	Rest	Floor Press ME	Speed Squat	Assisted leg/high repetition	Rest
Lunch	Agile drill	Shooting speed chain	Rest	Upper limb passing drill	The sprint direction changes	Race simulation	Rest

time/day	Mon (ME)	Tue (DE)	Wed	Thu (ME)	Fri (DE)	Sat (RE/GPP)	Sun
Night	GHR+ abdominal	triceps shoulder RE	Rest	Lat Pull + forearm	Calf/hip RE	Conditional Slide + Stretch	Rest

Week6

time/day	Mon (ME)	Tue (DE)	Wed	Thu(ME)	Fri (DE)	Sat (RE/GPP)	Sun
Morning	Good morning ME	Close Grip Bench	Rest	JM Press ME	Box Squat Speed	Posterior Chain Work	Rest
Lunch	Vertical jump pass	Spin throw	Rest	Shoulder stabilization	Jump sprint	Small confrontations	Rest
Night	Reverse Hyper core	biceps /Back RE	Rest	Tricep Ext Grip strength	Leg curls/ calves	Foam roller	Rest

Key Exercise Dosage Instructions：

Box Squat/Good Morning ME Variation: Rotate weekly, to 90-95% 1RM single rep, 1-3 reps, rest 3-5 min, only one main lift.

Bench/Floor Press DE/ME: DE: 9 sets x 3 reps @ 50-60% 1RM + 25% band/chain resistance, full speed; ME: To 90%+ single rep.

Speed Squat DE: 10-12 sets x 2 reps @ 55-60% + resistance band/chain, explosive lift, 45-60s rest.

GHR/Reverse Hyper RE: 4-5 sets x 10-15 reps, moderate weight, focus on posterior chain endurance.

Lat Pull/Tricep Ext RE: 3-4 sets x 12-20 reps, high reps to address weaknesses, using bands/dumbbells.

Abs/Grip Strength: 3 sets x 15-20 reps, variations like hanging leg raises.

Rotate primary lifts weekly to prevent adaptation—integrate netball-specific skills. DE days emphasise speed. Saturday's focus is on GPP endurance building. Total volume is moderate to avoid overtraining.

Adapted from: Lorenz et al. (2015); Schoenfeld et al. (2021); Goto et al. (2004)

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